



# **HET-CD-003**

## **(Power Processing Unit)**

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## Section 1 Component Scope

In order to provide the proper power loads to each of the hall effect thruster (HET) subsystems there needs to be a system to distribute those loads. This will be known as the power processing unit. This is commonly found on spacecraft to perform the task described previously.

### 1.1 Overview

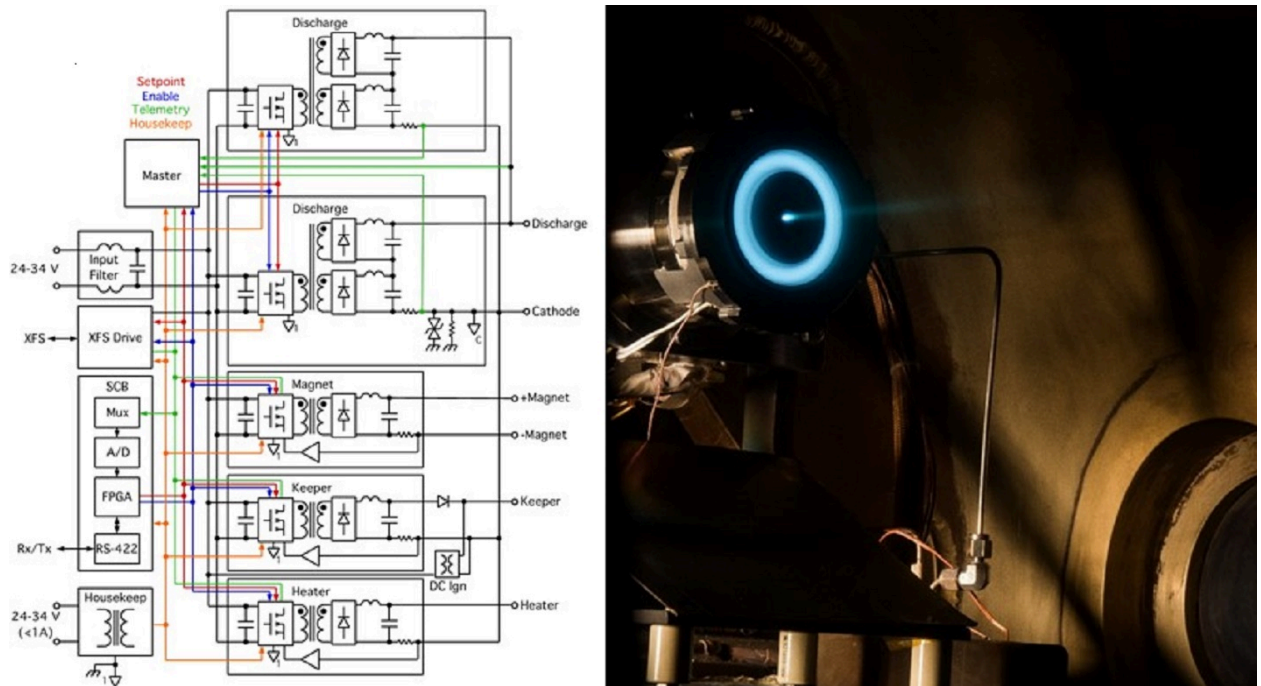
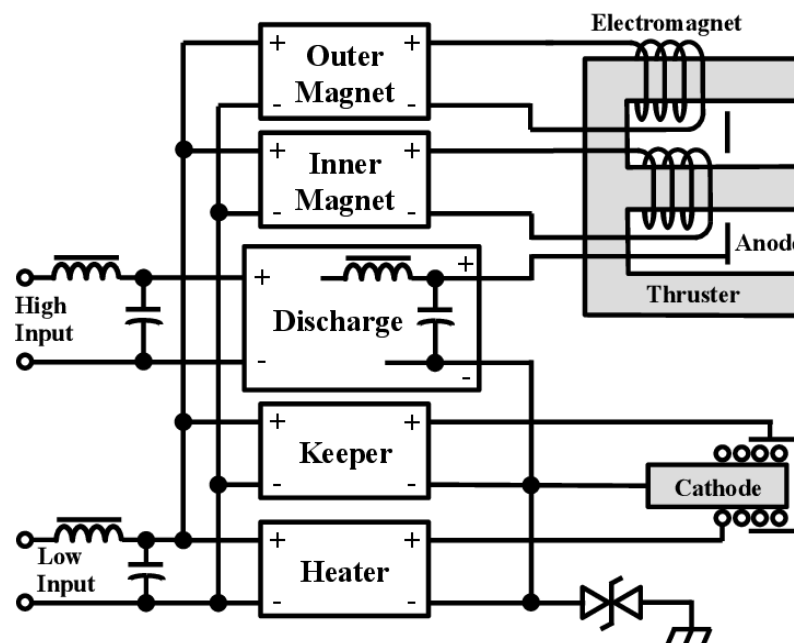


Figure 1: Example Diagram of PPU from NASA Low-Power HET



**Figure 2: Example of Simplified Block Diagram for PPU of High-Power HET**

## **1.2 Role in Assembly**

The Power Processing Unit (PPU) is the subsystem that distributes the power supply's output to subsystems within the HET. It moderates supply by stepping it up and down, and converts current between alternating and direct. The PPU distributes power to the following subsystems: ESP32 Board (and indirectly the Thermocouples and Load Cell), Cathode, Anode and Electromagnetic Coils. Each subsystem requires a specific range of voltage and power, which will be set by the PPU.

## **1.3 Design Objectives**

The key objective is to provide reliable and safe power loads to the subsystems. It must be able to withstand all thermal loads and stress/vibrations. It must be able to withstand the vacuum environment.

# **Section 2 Requirements and Constraints**

This section outlines the requirements and constraints necessary for the PPU.

## **2.1 Performance Requirements**

The input power will be X, and it will be distributed at X V and X Amps to each specific subsystem.

### **2.1.1 Primary discharge circuit**

The voltage and current requirements for the power supply that will be supplying the primary discharge current of the thruster is 1.06A at 375V. The plan is to overshoot these requirements by a smidge so as to give us a margin of error that we can use to tune the power supply to our needs. The circuit supplying power to the heater needs to be able to supply 50-100 Watts for a short period of time in order to heat the cathode to the needed temperature.

### **2.1.2 Coil Power supply**

The coils need to be supplied with 1 amp of current; the resistance of the coils is estimated at 1.68 ohms for all 4 of them in series (will probably change as the current simulation models progress.) This means the power supply will need to supply 1 Amp at 1.68 V. The supply into the PPU will have a therefore the supply will need to step this voltage down. The current circuit choice is a buck converter.

#### **2.1.4 Keeper supply**

The keeper supply will need to deliver a pulse of 1 Amp at around 600 Volts(The voltage is dependent on the gas flow rate). I will then have to switch to a steady state mode where it delivers tens of volts. The exact steady state current and voltage will be determined through testing.

## **2.2 Physical Constraints**

It must be able to be attached to the test stand, and have enough power connections to receive and distribute power.

## **2.3 Manufacturing Constraints**

The entirety of the PPU will be manufactured in-house from printed boards and soldered components. Components will be SMD and through hole, designed and selected with integration in mind via cross-team collaboration.

## **2.4 Environmental Constraints**

It must be able to survive inside the vacuum chamber such as the operating temperature, vibrations, shocks, and internal radiation inside the chamber.

## **2.5 Budgetary Constraints**

There is no budgetary constraint as of now.

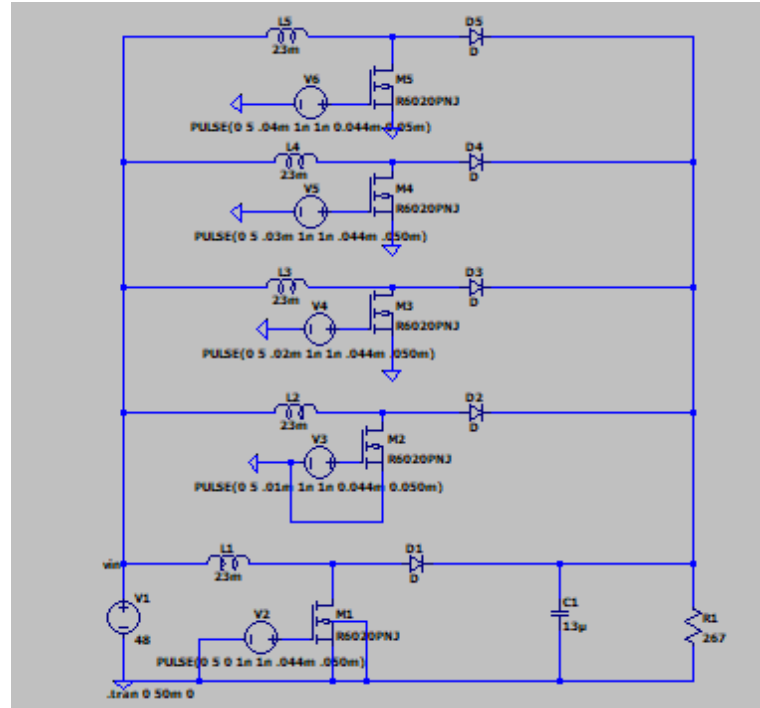
## **Section 3 System Design**

The preliminary planning will be done using napkin calculations to get the general idea for potential resistor, inductor, capacitor, and transformer values based on the required channel power output. It will then be simulated in LTspice and designed using Altium.

## **3.1 Geometry and Dimensions**

### **3.1.1 Primary Discharge Circuit**

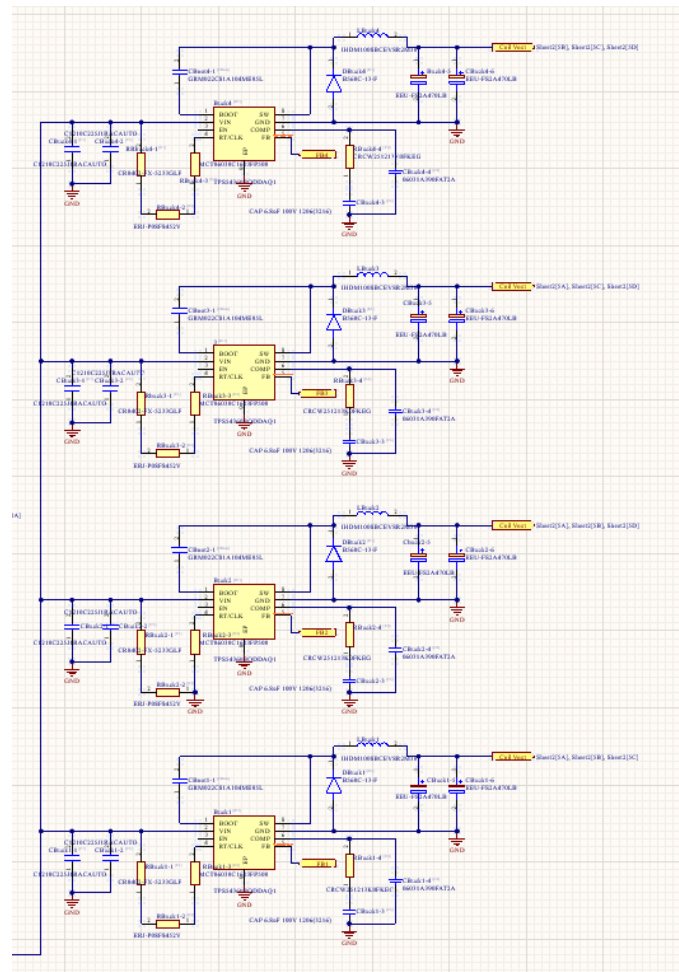
The primary discharge current will be supplied by a multiphase boost converter. A multiphase boost converter was chosen for its capability to supply the required output voltage and current at a high efficiency. It will consist of five phases; each phase will lead the previous phase by 1/5th of a period. The calculated control frequency is 20Khz. Each of the inductors is 23mH, themselves each composed of a series arrangement of 1x [631-RN122-4-02-3M3](#) 2x [631-RN122-2.5-02-5M6](#) 1x [631-RN122-3-02-4M5](#). The output capacitor is 13 uF. The mosfets are [R6020PNJs](#). They have a max Vds of 600V, Vgs of 30V, and a pulsed current of 80A.



**Figure 3: Primary Discharge Circuit**

Above is the circuit diagram for the primary discharge circuit. R1 is the simulated load. The load is expected to vary. The output voltage can be controlled by changing the duty cycle of the signals supplied to the mosfets. The control signals will be generated by the PPU's control unit. The control unit will have to account for changes in the plasma's resistance.

### 3.1.2 Magnetic Coil Circuit



**Figure 4: Magnetic Coil Power Circuit**

Above is the full circuit diagram for the circuit that will be powering the magnetic coils. It is a collection of four buck converters each feeding one of the four magnetic coils. Each buck converter can be controlled individually via their FB pin.

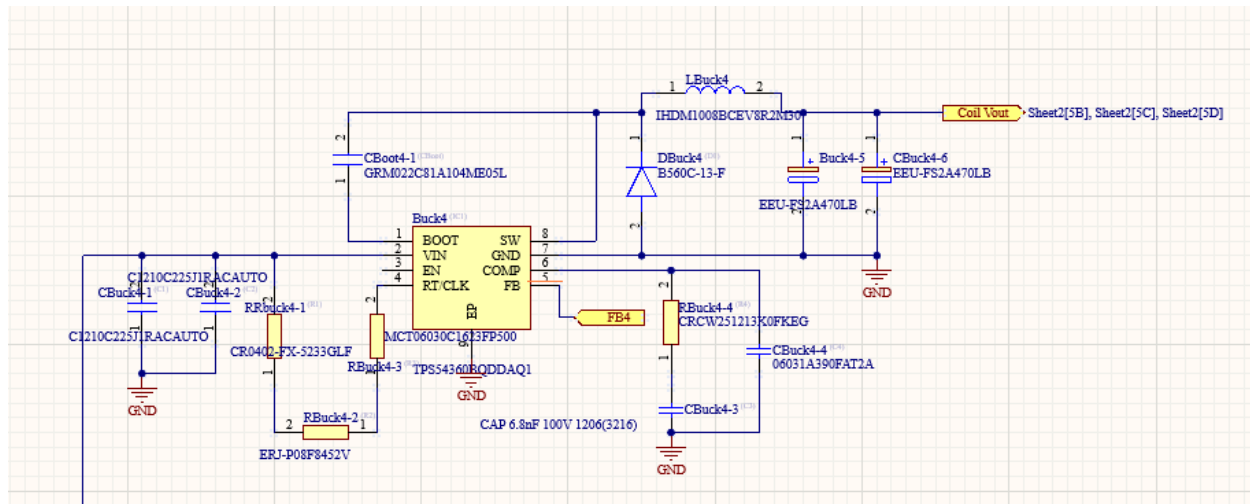


Figure 5: Magnetic Coil Power Circuit

Above is the individual circuit diagram for the buck converter. We are using the TPS54360BQDDA1

### 3.1.3 Keeper Circuit

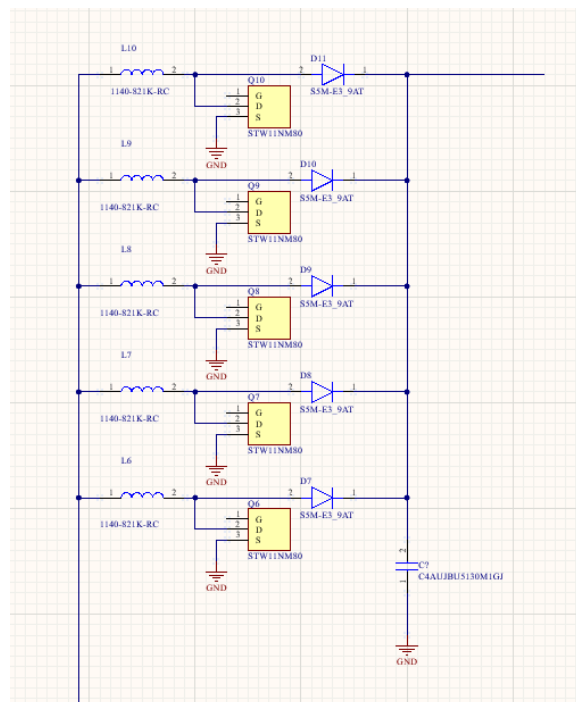


Figure 6: Keeper Power Circuit

The above keeper circuit is almost identical in operation to the primary discharge circuit. It takes five square wave signals as control inputs whose duty cycles are increased or decreased depending on the desired voltage.

### 3.2 Material Selection

List chosen material. Justify based on strength, thermal properties, machinability, and cost. Compare alternatives if relevant.

The material selection criteria is TBD.

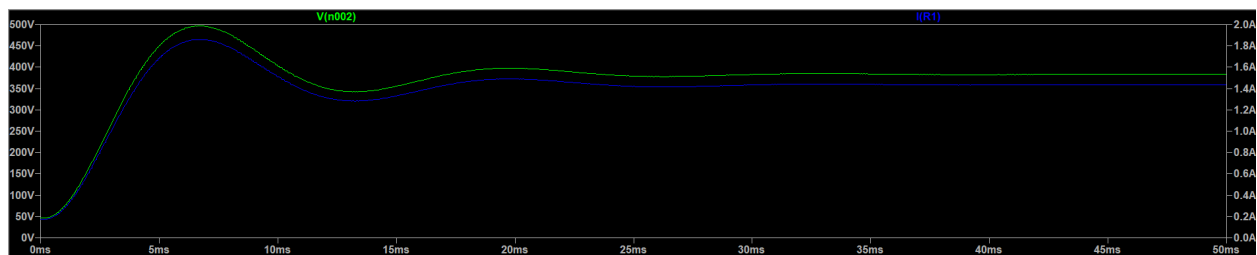
### 3.3 Interface Definition

Describe all interfaces (mechanical, electrical, thermal). Specify alignment, fasteners, and connections to other parts.

The interfaces are TBD.

### 3.4 Analysis and Validation

The discharge module of the power supply was simulated in LTSpice. It met the given specifications with an efficiency of %95.



## Section 4 Manufacturing Plan

An electrical schematic will be sent to a PCB manufacturer with all necessary dimensions, and component placements.

### 4.1 Fabrication Method

We will have the boards printed by a third party vendor and then we will assemble the components onto the board by means of a heat gun and soldering iron.

### 4.2 Facility Capabilities

JLPCB will print the boards and we can solder them wherever soldering irons and the accompanying tools can be found. Coren personally should have all of the equipment needed to assemble the board.

## 4.3 Inspection

## Section 5 Testing and Verification

Once the PCB is received by the club from the manufacturer, tests will be conducted using a multimeter and the ESP32 to determine functional code and proper power output loads from the subsystem channels.

### 5.1 Pre-Test Requirements

Proper PPE must be worn when handling the PPU as physical contact with one of the high voltage terminals may result in death. No part of a person's body should be uncovered if handling the PPU while power is being supplied. The PPU should be fully disconnected from all power sources before being moved or handled by a person. When powered no one should be within the vicinity of the PPU and the area around it should be free of any potential hazards.

### 5.2 Verification Tests

Ensure that all the power load output channels are functioning as intended (all subsystems need specific voltage and current values tested using an oscilloscope or other device). Test MCU and FPGA code to see if commands are properly sent and interpreted (start up sequence, open and closing of solenoids).

## Section 6 Safety and Risk Mitigation

### 6.1 Safety & Risk Mitigation

There is a high risk of electrocution and death. Always ensure that proper PPE is worn.

## Section 7 Cost Analysis

### 7.1 Bill of Materials

| Item              | Quantity | Total Price(unit price) | Link                                                                                                                                                                              |
|-------------------|----------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CR0402-FX-5233GLF | 1        | \$0.10                  | <a href="https://www.mouser.com/ProductDetail/Bourns/CR0402-FX-5233GLF?qs=sGAEpiMZZMtIubZb">https://www.mouser.com/ProductDetail/Bourns/CR0402-FX-5233GLF?qs=sGAEpiMZZMtIubZb</a> |

|                     |   |                |                                                                                                                                                                                                                                                                                 |
|---------------------|---|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     |   |                | <a href="#">dhIBIK1I%2F7FNVFNK5I<br/>sWdnnZ1Uw%3D</a>                                                                                                                                                                                                                           |
| ERJ-P08F8452V       | 1 | \$0.29         | <a href="https://www.mouser.com/ProductDetail/Panasonic/ERJ-P08F8452V?qs=sGAEpiMZZMtubZbdhIBIK7FbVYKWV6WVUxJQ35ttf0%3D">https://www.mouser.com/ProductDetail/Panasonic/ERJ-P08F8452V?qs=sGAEpiMZZMtubZbdhIBIK7FbVYKWV6WVUxJQ35ttf0%3D</a>                                       |
| MCT06030C1623FP500  | 1 | \$0.16         | <a href="http://mouser.com/ProductDetail/Vishay-Beyschlag/MCT06030C1623FP500?qs=sGAEpiMZZMtubZbdhIBIMi2voYIIRG89DAjm1Idels%3D">http://mouser.com/ProductDetail/Vishay-Beyschlag/MCT06030C1623FP500?qs=sGAEpiMZZMtubZbdhIBIMi2voYIIRG89DAjm1Idels%3D</a>                         |
| C1210C225J1RACAUT0  | 2 | \$1.04(\$0.52) | <a href="https://www.mouser.com/ProductDetail/KEMET/C1210C225J1RACAUT0?qs=s6nfgz3miVwiaTK1FL9dbA%3D%3D">https://www.mouser.com/ProductDetail/KEMET/C1210C225J1RACAUT0?qs=s6nfgz3miVwiaTK1FL9dbA%3D%3D</a>                                                                       |
| GRM022C81A104ME05L  | 1 | \$0.14         | <a href="https://www.mouser.com/ProductDetail/Murata-Electronics/GRM022C81A104ME05L?qs=sGAEpiMZZMsh%252B1woXyUXjwVqLNpRsvnQXE8lta5ZOF8%3D">https://www.mouser.com/ProductDetail/Murata-Electronics/GRM022C81A104ME05L?qs=sGAEpiMZZMsh%252B1woXyUXjwVqLNpRsvnQXE8lta5ZOF8%3D</a> |
| CRCW251213K0FKEG    | 1 | \$0.33         | <a href="https://www.mouser.com/ProductDetail/Vishay/CRCW251213K0FKEG?qs=sGAEpiMZZMtubZbdhIBIIF05JjrlZnUx73CQhI0Gjl%3D">https://www.mouser.com/ProductDetail/Vishay/CRCW251213K0FKEG?qs=sGAEpiMZZMtubZbdhIBIIF05JjrlZnUx73CQhI0Gjl%3D</a>                                       |
| C1608C0G2A682J080AC | 1 | \$0.20         | <a href="https://www.mouser.com/ProductDetail/TDK/C1608C0G2A682J080AC?qs=sGAEpiMZZMsh%252B1woXyUXj6cbJqqnAu83ua5i45P4HIU%3D">https://www.mouser.com/ProductDetail/TDK/C1608C0G2A682J080AC?qs=sGAEpiMZZMsh%252B1woXyUXj6cbJqqnAu83ua5i45P4HIU%3D</a>                             |
| 12061C682KAT2A      | 1 | \$0.30         | <a href="https://www.mouser.com/ProductDetail/KYOCERA-AVX/12061C682KA">https://www.mouser.com/ProductDetail/KYOCERA-AVX/12061C682KA</a>                                                                                                                                         |



## Section 8 References

| Document No. | Document Title | Date | Author |
|--------------|----------------|------|--------|
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<https://electricrocket.org/IEPC/IEPC-2009-117.pdf> - This is a great reference for what a PPU is and how power is distributed to the other subsystems. - Matthew Fan

<https://sqpcb.com/design-custom-circuit-boards/> - This should be a good introductory resource on how to design a PCB. - Max Day

## **Section 9 Appendices**

### **Appendix A –**